

1 Solving the Schrodinger equation

The infinite square well has a potential:

$$V = \begin{cases} \infty & x \leq 0 \\ a \geq x & \\ 0 & 0 \leq x \leq a \end{cases}$$

Thus inside the well $V = 0$ so the schrodinger equation reads:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = i\hbar \frac{\partial \Psi}{\partial t} = E\Psi$$

With the added constraint that $\psi(0) = \psi(a) = 0$ since beyond this region the infinite potential naturally makes ψ vanish. Our aim is to look for separable solutions $\Psi(x, t) = \psi(x)\eta(t)$

$$\frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2}\psi$$

looks exactly like the equation

$$\frac{d^2y}{dt^2} = -\omega^2 y$$

met in classical simple harmonic motion (SHM). The same equation means the same solution and so without integrating anything we can write:

$$\psi(x) = A \sin(kt - \phi)$$

for some constants, A , ϕ and $k = \sqrt{\frac{2mE}{\hbar}}$

To solve the rest of the problem we must find the constants A and ϕ and also find the time dependence of $\Psi(x, t)$, the wave-function of space and time. Recall $\psi(0) = 0$. Since $\sin(0) = 0$ we must have $\phi = \pi n$. Interestingly we can actually just set $\phi = 0$ here because if it is an even integer of π our wavefunction will be the same and if it is an odd integer of π the wavefunction will have a different sign, which does not matter since only the wavefunction's modulus squared will give us probabilities.

To find A we will "normalise" the wavefunction, which means ensuring that the total probability of finding a particle in the well is 1 as expected. Mathematically:

$$\begin{aligned}
\int_0^a \psi^2(x) dx &= 1 \\
A^2 \int_0^a \sin^2(kx) dx &= 1 \\
A^2 \left[\frac{kx}{2k} + \frac{\sin 2x}{4k} \right]_0^a & \\
\frac{A^2 a}{2} &= 1
\end{aligned}$$

Now recall $\psi(a) = 0$ so since $\sin(n\pi) = 0$, $ka = n\pi$.

$$A = \sqrt{\frac{2}{a}}$$

Thus we have:

$$\psi(x) = \sqrt{\frac{2}{a}} \sin(kx)$$

where the condition on k also gives us a discrete set of energies E we will label E_n :

$$\begin{aligned}
2mE_n/\hbar^2 &= k^2 = n^2\pi^2 \\
E_n &= \frac{\hbar^2\pi^2 n^2}{2m}
\end{aligned}$$

All that is left is to find time dependence. If we put subscripts on k like with energy so that $k_n = n\pi/a$ and also on our wavefunctions:

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin(k_n x)$$

we can write for $\Psi_n(x, t) = \psi_n(x)\eta_n(t)$

$$i\hbar \frac{\partial \Psi_n(x, t)}{\partial t} = E_n \Psi_n(x, t)$$

since the spatial part ψ can be taken out and cancelled.

$$i\hbar \frac{\partial \eta_n(t)}{\partial t} = E_n \eta(t)$$

so

$$\eta(t) = \eta(0)e^{-iE_nt/\hbar}$$

$$\Psi(x, t) = \sqrt{\frac{2}{a}} \sin(k_n x) \eta(0) e^{-iE_nt/\hbar}$$

Note that the complex conjugate of $e^{i\varphi}$ is $e^{-i\varphi}$ and they both multiply to one. Thus, $\eta(0) = 1$ to maintain normalisation. This has the interesting consequence of probabilities being time independent for single energy states: non trivial time dependence occurs if we take linear combinations of these states, which are equally valid solutions.

$$\Psi(x, t) = \sqrt{\frac{2}{a}} \sin(k_n x) e^{-iE_nt/\hbar}$$

2 Demonstrating the uncertainty principle

First we calculate $\langle x \rangle$ and $\langle x^2 \rangle$

$$\langle x \rangle = \int_0^a x \psi^2 dx$$

$$\langle x \rangle = \int_0^a \frac{2}{a} x \sin^2 kx dx$$

We use integration by parts:

$$\frac{a\langle x \rangle}{2} = \left[\frac{x^2}{2} - \frac{\sin(2kx)}{4k} \right]_0^a - \int_0^a \frac{x}{2} - \frac{\sin(2kx)}{4k} dx$$

$$\frac{a\langle x \rangle}{2} = \left[\frac{x^2}{2} - \frac{\sin(2kx)}{4k} - \frac{x^2}{4} + \frac{\cos(2kx)}{8k^2} \right]_0^a$$

recall that $ka = n\pi$ and that $\cos 2\pi n = 1$ and $\sin 2\pi n = 0$ so:

$$\langle x \rangle = \frac{2}{a} \left[\frac{a^2}{2} - \frac{a^2}{4} \right] = \frac{a}{2}$$

$$\langle x^2 \rangle = \int_0^a \frac{2}{a} x^2 \sin^2 kx dx$$

$$\begin{aligned}
&= \frac{2}{a} \left[\frac{x^3}{2} - \frac{x^2 \sin 2kx}{4k} - \frac{x^3}{2} - \frac{x \cos 2kx}{4k^2} + \frac{x^3}{6} + \frac{\sin 2kx}{8k^3} \right]_0^a \\
&= a^2 \left(\frac{1}{3} - \frac{1}{2n^2\pi^2} \right)
\end{aligned}$$

The standard deviation will be:

$$\begin{aligned}
\sigma_x &= \sqrt{\langle x^2 \rangle - \langle x \rangle^2} \\
&= a \sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}}
\end{aligned}$$

Now for momentum:

$$\langle p \rangle = 0$$

and from the de-Broglie wavelength relation $p = h/\lambda = \hbar k$ so:

$$\langle p^2 \rangle = \hbar^2 k^2 = \frac{\hbar^2 n^2 \pi^2}{a^2}$$

Finally Heisenberg's uncertainty principle can be tested:

$$\begin{aligned}
\sigma_p \sigma_x &= \hbar n \pi \sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}} \\
&= \frac{\hbar}{2} \sqrt{\frac{n^2 \pi^2}{3} - 2}
\end{aligned}$$

the minimum uncertainty is in the ground state $n = 1$ so:

$$\text{Min}(\sigma_p \sigma_x) = \frac{\hbar}{2} \sqrt{\frac{\pi^2 - 6}{3}}$$

But $\pi > 3$ and so $\pi^2 > 9$ meaning $\frac{\pi^2 - 6}{3} > 1$ and finally

$$\sigma_p \sigma_x > \frac{\hbar}{2}$$

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- 4 The finite square well in three dimensions?
- 5 Applications